GLENCORE

Master Data Standard

Work Centre

12

Document properties

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Department, Area	Maintenance, Glencore Copper
Prepared by	Ben Hathaway
Reviewed by	Regional PM SME's, Regional AM & SAP PM Managers
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1. Introduction

1.1 Master Data Standards - Purpose

The Data Standard provides clear and prescriptive definitions and rules that support a consistent and common definition of data entities across all business systems.

It describes:

- What rules / guidelines are applied to determine how many of the relevant data entity will be created.
- What rules will be applied to determine the codes used to create the relevant data entity (e.g. Plant code or Material code);
- What rules will be applied to determine the name / short description of the relevant data entity (e.g. Plant name or Vendor name);
- Rules for the use of views / sub-entities within the Data Entity;
- Identification of the key Attributes on the data entity whose usage needs to be standardised;
- When multiple values are available for the nominated 'key' attributes, the rules definition should list the criteria to use to determine the appropriate value; and
- If applicable, any rules / guidelines for the association of the relevant data entity with other data entities.

The Data Standard is **NOT** used to document:

- 'Technical' documentation on how the system has been configured and the design rationale for the configuration
- Identifying the operational need on when these Standards should be applied, They are to provide the guidance on how the Standard is applied
- List of actual values for the data entity / attributes within the data entity except where it is required to explain the application of the data standard
- User training or procedural documentation for master data maintenance
- Data migration

1.2 Work Centre Standard – Purpose

The purpose of this Standard is to:

- Identify the minimum Master Data field requirements for SAP Work Centres at Glencore.
- Ensure alignment with Solution Building Block SBB_D90 Maintenance Master Data and Asset Management Framework Element 5
- Identify the minimum requirements for creating, maintaining, and retiring the master data
- Provide the guiding principles to manage Work Centre master data
- Identify the primary audience for the application of this standard

1.3 Using this document

The Data Standard is targeted at the following audience:

- Master Data maintenance teams (IT and Business) in order to understand how the object needs to be maintained
- End users supporting the maintenance function whereby appropriate knowledge is required to use the system effectively which include but are not limited to: Planners, Schedulers, Reliability Engineers etc.
- Business Intelligence team to assist with the design of reporting solutions
- Project teams whose scope impacts the data object to ensure the solution design is compliant with the approved data standard
- Transition to operations teams to assist in the developing of user guides

1.4 Governance

Master Data Governance (MDG) is the governance tool used to create, maintain and retire Master Data objects in Plant Maintenance. The Asset Management group provides the governance for all requirements described in this Standard as well as approval of any change to related Master Data.

This Standard shall be reviewed as often as the business requires but at a minimum of every five years.

The Standard is subject to Glencore's Continuous Improvement (CI) processes based on Glencore's shared experiences and Lessons Learned.

Revisions, reviews, and updates to the Standard take place under the guidance and approval of Regional Asset Management and Subject Matter Expert (SME) leaders

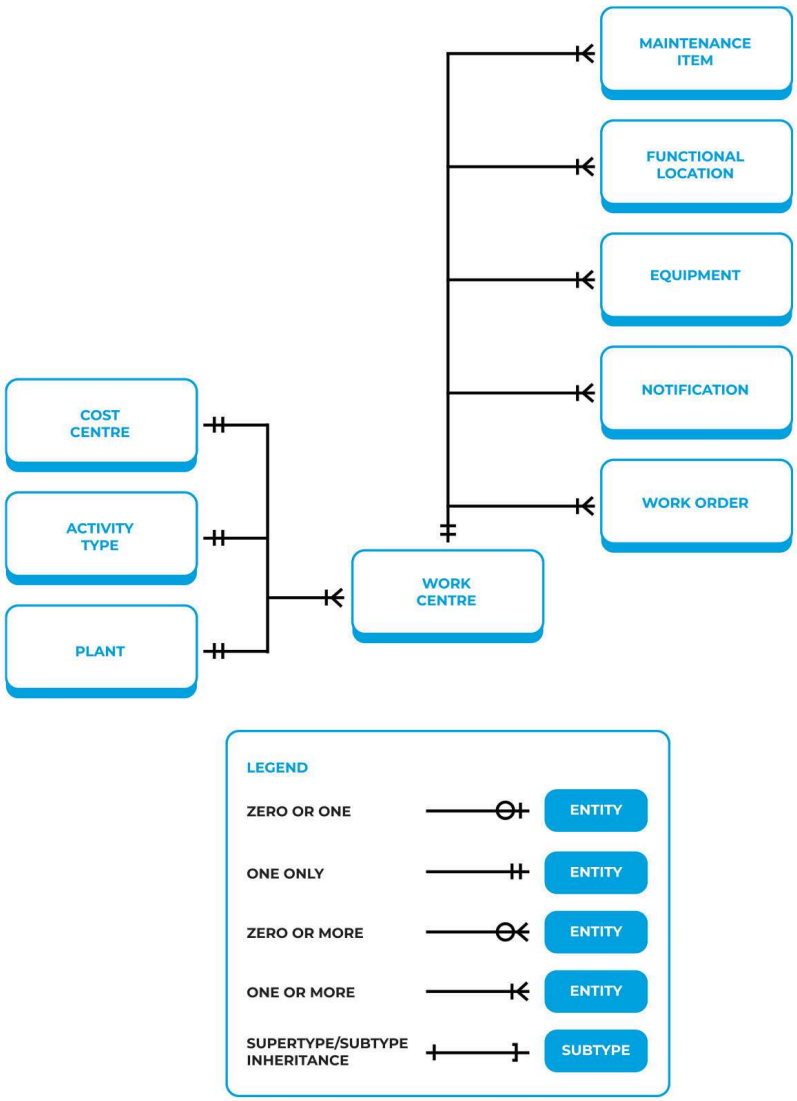
2. Data Object Description

2.1 Definition

A Work Centre is an organisational unit that defines by whom or what a maintenance task is performed.

The Work Centre has a capacity to do work and the activities performed by them can be costed by setting rates for those activities. A Work Centre can represent both people and machines/objects that need to be managed and or costed.

2.2 Entity Relationship Diagram (ERD)



3. Business Requirements

3.1 Business Use

3.1.1 Work Centre Types

Two types of Work Centres will be defined for use in Plant Maintenance processes: Main Work Centre and Resource Work Centre.

- Main Work Centres are used at a higher level and represent the group responsible for work management activities. They are assigned to Functional Locations and Equipment in the Technical Object Structure as well as the header of Work Orders, Task Lists and Maintenance Items.
- Resource Work Centres are used to define trade or skill types as well as physical resources and are used at the operation level on Work Orders and Task Lists.

Resource Work Centres include:

- Internal Labour – for labour resources that contain capacity, costing and scheduling data. May include both Employees and contractors.
- Contractor Labour – for contractor resources that require different capacity, costing and scheduling data to Internal Labour work centres and are planned as a separate operation on a work order. Not for activities performed and paid for as individual services on the work order.
- External Services – used when a specific Service or Equipment Hire is required. The Service or Hire is specific to a work order and as they are external resources, they are not capacity managed.
- Machines – Any finite resource that is integral to the execution of the planned work should be capacity managed and as such should have a Work Centre created. e.g. Cranes.

3.2 Business design Rules & Criteria

3.2.1 General

All Resource Work Centres will have a superior Main Work Centre.

3.2.2 Work Centre Cost Centres and Costing

All Work Centres will be assigned a Cost Centre to enable the Responsibility Key Reporting Area to be derived. Note that only Resource Work Centres will have a default activity and activity rate assigned.

Main Work Centres will not be costed.

3.2.3 Scheduling Capacities

Available capacity derived from intervals and shifts within a Work Centre Capacity will be utilised to manage changing shifts/capacity levels.

4. Detail Standard

4.1 Overview

There will be several differences in the setup and functionality of Main and Resource Work Centres in the system and this will assist in governing their use. Main Work Centres will not be used in capacity planning and will have no activity rates against them. As such, they will only be used at the header level on Work Orders, Task Lists and Maintenance Items as well as assigned to Functional Locations and Equipment in the structure to define areas of responsibility.

Resource Work Centres will have capacity and cost information assigned and will be used at the operation level of Task Lists and Work Orders and will be used for scheduling, capacity planning and costing of the Work Orders.

4.2 Data Object Coding Standard

To ensure a visible distinction between Main Work Centres and Resource Work Centres, the length shall be different. As Resource Work Centres will report to a Main Work Centre, a reference to the Main Work Centre is included in the name of the Resource Work Centre.

The table below is an example (not a full list) for the naming of Work Centres used for Australian (AU) Queensland Metals business configuration. See Data Standard – General Configuration for further information.

Work Centre number	Description	Value
Characters 1 - 2 (MWC & RWC)	Location	EH - Ernest Henry MI - Mount Isa CS - Copper Smelter GF - George Fisher
Characters 3 - 4 (MWC & RWC)	Department	CN - Concentrator UG - Underground Mine MF - Mobile Fleet Workshop SO - Surface Operations
Characters 5 - 6 (MWC & RWC)	Discipline	UO - Underground Operations MO - Mobile Maintenance LU - Lubrication LV - Light Vehicle
Characters 7 – 8 (RWC Only)	Trade	EN - Engineer DF - Diesel Fitter TF – Tyre Fitter BM - Boilermaker

Work Centre naming from a previous legacy system may in some instances have been carried across to SAP and this Standrad does not look to make retrospective change. However, any additional SAP Phase upgrades or new installations shall allow for the development of a naming standard suitable for that Plant that provides clear instruction. This information shall be made available in the General Configuration tables.

4.3 Attributes

4.3.1 Work Centre Control Key

Application Definition	The Code or Key which determines how the Work Centre will be used. Enables Scheduling, internal and external costing.
Guiding Principles	The following Control Keys shall be used PM01 – Plant Maintenance – Internal PM02 – Plant Maintenance – External – Subcontract PM03 – Plant Maintenance – External (Services) PM04 – Plant Maintenance – Internal (Service)
Application Coding Convention	Coding = 4 character alphanumeric
Naming Convention	Refer to Guiding Principles above
Reporting Considerations	Used in the reporting of Costs and Maintenance History of technical objects

4.4 Hierarchy Standards

Work Centres will have an implicit hierarchy based on the naming convention of the Work Centre.

The Main Work Centre code always comprises six characters.

The Resource Work Centre code comprises eight characters. The first six characters represent the Main Work Centre that is responsible for the Resource Work Centre.

5. APPENDIX A: ADDITIONAL INFORMATION

5.1 Complete Field List

Legend for Attribute Table

- **Table –** SAP table where data object appears
- **Field –** SAP data attribute field
- **Field Name –** SAP field name of data object
- **Description –** Description of input required
- **Data Type –** SAP data type (CHAR=Character, DATS=Date, TIMS=Time, UNIT=Measurement type, NUMC=Number)
- **Length –** Character length of field
- **MOC –** Data requirement (M=Mandatory, O=Optional, C=Conditionally mandatory)
- **Field Validation Type –** Entry type allowed (e.g. Free text, List selection) and additional information

Table	Field	Field Name	Description	Data Type	Len	MOC	Field Validation Type
P3000	WERKS	Plant	Maintenance Plant defined uniquely with a four numeric code	CHAR	4	M	Table Select: Valid list of maintenance plants
P3000	ARBPL	Work Centre	Key Identifying the work centre	CHAR	8	M	See Data Standard – General Configuration
P1000	STEXT	Description	The Work Centre header description. The text must contain a succinct, consistent and informative description.	CHAR	40	M	See Data Standard – General Configuration
P3000	VERWE	Category	Key which distinguishes the Work Centres by their category	CHAR	4	M	Table Select: Work Centre Category 0005 - Plant maintenance only
P3000	VERAN	Person Responsible	The person or group of people who are responsible for the maintenance of the work centre	CHAR	3	M	001 – Supervisor Mechanical 002 – Supervisor Electrical 003 – Supervisor Project 004 – Supervisor Project
P3000	PLANV	Usage	Key to control which type of task list you can use: operating resources (for example, Work Centres or production resources/tools)	CHAR	3	M	Table Select: 004 – Only maintenance task lists
P3000	VGWTS	Standard Value Key	Key defining and giving a dimension (for example, time or area) to one of up to six standard values.	CHAR	4	M	Table Select: SAP0 - No Standard values
P3003	STEUS	Control Key	Determines which business transactions should be executed for the object that belongs to the task list or order (for example scheduling or costing)	CHAR	4	C	For Resource Work Centres only: Table Select: PM01 – Plant Maintenance – Internal PM02 – Plant Maintenance – External – Subcontract

Table	Field	Field Name	Description	Data Type	Len	MOC	Field Validation Type
							PM03 – Plant Maintenance – External (Services) PM04 – Plant Maintenance – Internal (Service)
RC68A	KAPART	Capacity Category	Key which describes capacity in more detail. The capacity category enables you to determine whether it is machine or labour capacity.	CHAR	3	C	For Resource Work Centres only: Table Select: 001 – machine for non-human Work Centres 002 – person for human resource Work Centres
P3006	FORKN	Other formula	Formula used to calculate capacity requirements for other types of internal processing (for example, in maintenance orders)	CHAR	6	C	For Resource Work Centres only: Table Select: SAP008 Only
KAKO	PLANR	Capacity Planner Group	The capacity planner group is configured independently of plant and is used to manage capacities.	CHAR	3	C	For Resource Work Centres only: Table Select: A11 – AU Mech Planner A12 – AU Elect Planner A13 – AU Service Planner A14 – AU Production Planner
KAKO	KALID	Factory Calendar	The factory calendar when used with available capacity and shift sequence, determines the available capacity by week taking into account non-working days, public	CHAR	2	C	For Resource Work Centres only: Table Select: valid value for your Plant

Table	Field	Field Name	Description	Data Type	Len	MOC	Field Validation Type
			holidays etc. for normal available capacity within a Work Centre.				Default - Z7 – 7 Working Days
KAKO	MEINS	Unit of Measure	The resource capacities unit of measure is normally hours (H)	UNIT	3	M	Table select: H for hours
RC68K	BEGZT	Start Time	The normal start time for the Work Centre capacity within a shift	TIMS	6	C	For Resource Work Centres only:
RC68K	ENDZT	Finish Time	The normal end time for the Work Centre capacity within a shift.	TIMS	6	C	For Resource Work Centres only:
RC68K	PAUSE	Length of Break	The main break time for a normal shift	TIMS	6	C	For Resource Work Centres only: Enter the duration of the shift's main break
KAKO	NGRAD	Capacity Utilisation	The utilisation percentage of the capacity within the Work Centre	NUMS	3	C	For Resource Work Centres only:
KAKO	AZNOR	Number of Individual Capacities	The discrete quantity of each capacity within the Work Centre	INT2	5	C	For Resource Work Centres only: Enter the total quantity of the capacity for the Work Centre.
KAKO	KAPTER	Indicator – relative to finite scheduling	An indicator that classifies the Work Centre as relative to scheduling	CHAR	1	C	For Resource Work Centres only: Always select for an operation resource Work Centre that shall be scheduled.

Table	Field	Field Name	Description	Data Type	Len	MOC	Field Validation Type
KAKO	KAPAVO	Can be used by several operations	You use this indicator to specify that several operations may use the available capacity of this capacity category. If you do not set the indicator, no further operations may access the remaining available capacity if an operation already requires a proportion of the available capacity	CHAR	1	C	For Resource Work Centres only: Always select for an operation resource Work Centre that shall be scheduled.
KAKO	UEBERLAST	Overload	Entry you use to control the percentage by which the load may exceed available capacity. If you have entered a percentage for the overload, it is also taken into account in the availability check for capacity. For example, if there is an overload of 10%, Work Centres with a capacity load of more than 110% are overloaded	NUMC	3	O	Overload should be 0%
RC68K	KAPLPL	Indicator – long term planning	Indicates that this capacity is taken into account at the time of long-term planning. The goal of long-term planning is to determine material requirements and capacities for assemblies and raw materials based on a simulated demand program containing independent requirements	CHAR	1	M	Indicator should be checked.
P3005	FORTN	Internal processing Duration	Scheduling – execution time.	CHAR	6	M	SAP004 - Proj:Netw/Maint.Time Uses duration (field DAUNO)
P3005	VGDIM	Dimension of Work	For hour based resource capacities the dimension of work shall be time.	CHAR	6	M	Always enter: TIME
P3005	VGARB	Unit of measure of work	The time unit measurement	UNIT	3	M	Always enter: H

Table	Field	Field Name	Description	Data Type	Len	MOC	Field Validation Type
P1001	BEGDA	Start Date	The effective date when the Work Centre shall first be applied to resource planning and scheduling	DATS	8	C	For Resource Work Centres only: Enter Start Date of Work Centre in Date Format: DD.MM.YYYY
P1001	ENDDA	End Date	Contains a date at that identifies the end of the validity period	DATS	10	C	For Resource Work Centres only: Enter Start Date of Work Centre in Date Format: DD.MM.YYYY Default 31.12.9999
CRKEYK	KOKRS	Controlling Area	Uniquely identifies a controlling area. The controlling area is the highest organisational unit.	CHAR	4	M	Always enter GCGC – Glencore Copper
CRKEYK	KOSTL	Cost Centre	The responsibility cost centre applicable to the Work Centre (from the person responsible)	CHAR	10	M	Table Select from Plant list: Select valid cost centre
PAD011	LARN1	Activity Type	The activity type linked to the responsibility cost centre valid with a planned rate for the current period	CHAR	6	C	For Resource Work Centres that will be costed Table Select: Select valid activity
PAD011	FORN1	Formula key for costing	Key which refers to the formula used to determine costs per activity type	CHAR	8	M	Enter value: SAP008 – Uses Work

6. Stakeholder Signoff

Stakeholder Name	Role	Contact	Stakeholder Sign-off
Lawrence Hlatswayo	Regional Manager – Asset Management & SAP PM Business Process (AFR)	Lawrence.Hlatswayo@glencore.co.za	
Anthony Green	Regional Manager – Asset Management & SAP PM Business Process (AU)	anthony.green@glencore.com.au	
Karl Nicolosi	Regional PM SME (AU)	karl.nicolosi@glencore.com.au	
Akeem Araoye	Regional Manager – Asset Management & SAP PM Business Process (NA)	Akeem.Araoye@glencore.ca	
Jean-Yves Beaulieu	Maintenance Systems Coordinator (NA)	Jean-Yves.Beaulieu@glencore.ca	
Yuri Yurac	Regional Manager – Asset Management & SAP PM Business Process (SA)	yuri.yurac@glencore.cl	
Erick Vicencio	Regional PM SME (SA)	Erick.Vicencio@glencore.cl	
Ghislain Bisson	Global SAP Business Process Owner	Ghislain.Bisson@glencore.ca	
Dino Moll	Senior Manager - Asset Management	Dino.Moll@glencore.com.au	

The Regional Team would like to acknowledge the efforts of the AM Network and Australian SAP PM Business Support Teams in the development of the Master Data Design Standards.

Anthony Green
Regional Manager – Asset Management & SAP PM Business Process
Glencore Copper Operational Excellence
M: +61 438907225
anthony.green@glencore.com.au
160 Ann St, Brisbane Qld 4000